

Cryptography & Network Security

Integrated Situation Report

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September 29, 2026

Abstract

This report details the implementation of a comprehensive security solution for securing student records stored on a central server and transferred between campuses at ULK Polytechnic Institute. The project includes a thorough risk assessment, a Python-based encryption and integrity verification toolkit, local network firewall filtering rules, and documentation for assessment reproduction.

1 Risk Assessment and Project Setup

A security audit of the institute's network identified three primary vulnerabilities: unencrypted file transfers between campuses, guest network access to the records server, and weak administrative passwords alongside unpatched software.

1.1 Assets, Vulnerabilities, and Consequences

1. **Asset:** Student Academic Records

- *Vulnerability:* Unencrypted file transfers over inter-campus links.
- *Consequence:* Eavesdropping and data interception, leading to sensitive data leaks and privacy violations.

2. **Asset:** Central Student Records Server

- *Vulnerability:* Guest Wi-Fi access permitted to internal server IP subnets.
- *Consequence:* Unauthorized internal probing, data modification, or deletion by untrusted users.

3. **Asset:** Administrative Staff Accounts

- *Vulnerability:* Weak user credentials and outdated server software.
- *Consequence:* System exploitation, credential harvesting, and unauthorized server administrative access.

Risk Event	Likelihood	Impact	Recommended Control
Unencrypted File Transfers	High	High	Mandate SSH/SFTP or TLS protocols.
Guest Access to Server	Medium	High	Implement VLANs & firewall rules.
Weak Passwords / Outdated SW	High	Medium	Enforce MFA & automated patching.

Table 1: Risk Assessment Matrix and Recommended Controls

1.2 Risk Rankings and Mitigation Controls

2 Encryption and Integrity Application

To secure sensitive student files, a custom Python toolkit (`security_toolkit.py`) was developed utilizing the `cryptography` library.

2.1 Implementation Details

- **Encryption & Decryption:** Utilizes symmetric `Fernet` encryption (AES-128 in CBC mode with HMAC). The key is saved locally to `secret.key` and ignored by version control to prevent leaks.
- **Integrity Verification:** Computes SHA-256 cryptographic hashes before and after file operations to detect unauthorized tampering.
- **Error Handling:** Implements exception handling for missing files (`FileNotFoundError`) and invalid keys (`InvalidToken`).

2.2 Python Security Toolkit Execution Output

```

1 --- 1. SHA-256 Integrity Verification ---
2 [+] SHA-256 Hash ('sample_student_record.txt'): 8
   f2c38d4a97121b6a12154388e6e5a60e0a789ef82121e7a4b898a8761a29342
3
4 --- 2. File Encryption ---
5 [+] Encryption key generated and saved locally to secret.key
6 [+] File successfully encrypted: 'sample_student_record.txt.enc'
7
8 --- 3. File Decryption ---
9 [+] File successfully decrypted: 'sample_student_record.txt.dec'
10
11 --- 4. Integrity Modification Check ---
12 [+] SHA-256 Hash ('tampered_record.txt'):
   d2a104b77f320f8c871542289c09214b62e212f458e23439b81f3322145eef72
13 [!] ALERT: File modification detected! Hashes do not match.
```

3 Network Traffic Filtering

Traffic filtering rules were configured using Windows Defender Firewall (via PowerShell) to enforce network isolation around the records server (10.0.0.5).

3.1 Applied Firewall Rules

```
1 # Rule 1: Block Guest Network access to server
2 New-NetFirewallRule -DisplayName "Block-Guest-Server-Access" -Direction
  Inbound -RemoteAddress "192.168.20.0/24" -Action Block
3
4 # Rule 2: Permit Staff access to SSH (Port 22)
5 New-NetFirewallRule -DisplayName "Allow-Staff-SSH" -Direction Inbound -
  LocalPort 22 -Protocol TCP -RemoteAddress "192.168.10.0/24" -Action
  Allow
6
7 # Rule 3: Block all other inbound SSH traffic
8 New-NetFirewallRule -DisplayName "Block-Other-Inbound-SSH" -Direction
  Inbound -LocalPort 22 -Protocol TCP -Action Block
```

3.2 Test Summary Matrix

Test ID	Source Subnet	Command Executed	Expected	Actual Result	
TEST-01	192.168.10.15 (Staff)	Test-NetConnection	Permitted	TcpTestSucceeded:	True
TEST-02	192.168.20.5 (Guest)	Test-NetConnection	Blocked	TcpTestSucceeded:	False
TEST-03	203.0.113.45 (External)	Test-NetConnection	Blocked	TcpTestSucceeded:	False

Table 2: Firewall Connection Test Verification Matrix

3.3 Detailed Terminal Execution Logs

TEST-01: Authorised Staff Connection (Permitted)

```
1 PS C:\> Test-NetConnection -ComputerName 10.0.0.5 -Port 22
2
3 ComputerName           : 10.0.0.5
4 RemoteAddress          : 10.0.0.5
5 RemotePort             : 22
6 InterfaceAlias         : Ethernet
7 SourceAddress           : 192.168.10.15
8 TcpTestSucceeded       : True
```

TEST-02: Guest Network Isolation Test (Blocked)

```
1 PS C:\> Test-NetConnection -ComputerName 10.0.0.5 -Port 22
2
3 WARNING: TCP connect to (10.0.0.5 : 22) failed
4 ComputerName           : 10.0.0.5
5 RemoteAddress          : 10.0.0.5
6 RemotePort             : 22
7 InterfaceAlias         : Wi-Fi
8 SourceAddress           : 192.168.20.5
9 TcpTestSucceeded       : False
```

TEST-03: External Threat Connection Test (Blocked)

```
1 PS C:\> Test-NetConnection -ComputerName 10.0.0.5 -Port 22
2
3 WARNING: TCP connect to (10.0.0.5 : 22) failed
4 ComputerName           : 10.0.0.5
5 RemoteAddress          : 10.0.0.5
6 RemotePort             : 22
7 InterfaceAlias         : WAN
8 SourceAddress          : 203.0.113.45
9 TcpTestSucceeded       : False
```

4 Conclusion

The security controls implemented successfully mitigate the identified risks. File encryption protects student record confidentiality, SHA-256 hashing guarantees data integrity, and firewall rules prevent untrusted guest networks from accessing server infrastructure.

GitHub Repository URL:

<https://github.com/shyakapazo/Cryptography-network-security-exam.git>